

State of Knowledge — Fri–Sun 2026-06-06/07
weekend consolidation. The mass=volume
spine, organized by tier (DERIVED /
CANDIDATE / SCAFFOLD / RETRACTED /
OPEN), for the team’s next-steps reconvene.

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State of Knowledge — Weekend 2026-06-06/07

Casey: “document what we know, then come back.” This is the honest map. The weekend’s shape: the substrate handed back a shelf; the team tested every board at velocity; a small load-bearing core survived and got sharper; several elegant extensions went on the floor under test. What follows is organized strictly by tier — derived stands, candidate is flagged, retracted is recorded so we don’t rebuild it.

1. The spine (one paragraph)

Mass is the substrate volume an object occupies. A particle winds the n_C -complex-dimensional substrate D_{IV}^5 ; its mass is how much of that volume it wraps, in units of $\pi^{n_C} = \pi^5$ (the flat volume exponent of the substrate’s own complex dimension). The five dimensions don’t appear as space we move through — they appear as weight. How *much* you wind is mass (the π^5 , conformal- ρ /volume side); *whether the winding band twists* is boson-vs-fermion (the Z_2 , compact- ρ /spectral side). One act of winding, two visible faces.

2. DERIVED (solid — build on these)

- $\pi^5 = \pi^{n_C}$ (F52; Elie Toy 4015; Grace catalog). The flat Euclidean volume exponent of the $n_C=5$ complex-dimensional substrate (real dim $2n_C=10$; $\text{Vol}(B^{2n_C}) = \pi^{n_C}/n_C!$). Three independent routes converged. Shilov-winding (π^3) ruled out.
- $m_p/m_e = C_2 \cdot \pi^{n_C} = 6\pi^5$ at 0.002%; the universal π^5 factors out of all real cross-masses.

- Measure role-split (F54): masses ride the *flat* volume $\pi^{\{n_C\}}$; the FK *invariant* measure is $\pi^{\{n_C-1/\text{rank}\}} = \pi^{\{9/2\}}$ ($c_{\text{FK}} \cdot \pi^{\{9/2\}} = 225$). Gap = $1/\text{rank}$, a half-integer Γ -shift. Flat = extensive (mass); invariant = spectral (inner product).
- $R(k) = C(k,2)/\kappa_{\text{Bergman}}$ (F47/F49; Elie 23/23, zero conformal offset, $k=2..24$). Heat-trace sum-of-roots = binomial over Bergman curvature $\kappa_{\text{Bergman}} = -n_C$. Uses the compact K-Casimir $\rho_{\text{SO}(5)} = (3/2, 1/2)$.
- Phase 3 integers, both derived: vector mesons (ρ, ω) wrap $n_C = 5$ (dim of the vector rep $V_{-}(1,0)$); baryons (p, n) wrap $n_C + 1 = 6 = C_2$ (the Z_2 spinor twist — derived four independent ways: Lyra cells $= n_C + 2\lambda_2$, Keeper $H^1(Z_2)$ cohomology, Elie Möbius 4π , Grace $|Z_2|=2$). The +1 is forced (order of $Z_2 = T185$, registered), not a fit.
- The real cross-mass set is 4 light u/d hadrons (p, n, ρ, ω) — Elie/Grace base-rate filter; the inflated “~200” were dev%-flatters-large-numbers illusions (Y’s “60” is 60.50, $p=0.99$).

3. CANDIDATE (strong, on standing principles — flagged, not closed)

- Mass = the substrate beyond the conformal group $\text{SO}(4,2)$ (F53, on Casey #14 STANDING). The chain $\text{SO}(5,2) \rightarrow \text{SO}(4,2) \rightarrow \text{SO}(3,1)$: $\text{SO}(4,2)$ is the conformal group of a *massless* 3+1 world; mass is the residual $\text{SO}(5,2)/\text{SO}(4,2)$ breaking scale invariance. How the 5D shows in 3+1: as the existence of mass.
- cells $= n_C + 2\lambda_2$ (F57). The spinor-geometry law: bulk wrap n_C + the doubled phase-circle winding $2\lambda_2$ (the Dirac 4π double cover). Predictive (not a 2-point fit): $\lambda_2=1 \rightarrow 7\pi^5$ (1095 MeV), $\lambda_2=3/2 \rightarrow 8\pi^5$ (1251 MeV). The genuinely-next testable program.
- Dual- ρ structure (F49), now *asymmetric* (Grace): compact $\rho_{\text{SO}(5)}$ (spectral, π^2 , “measures”) vs conformal ρ (Plancherel/volume, π^5 , “adds mass”). The π -powers split by ρ -role. Leptons spectral point-modes (π^2); hadrons bulk-volume (π^5).
- Reading (a) (F44): everything physical lives in H^2 ; no physical $(1-P)$ complement. Forecloses the Cal #259 conflation; the foundation for the volume picture.

4. SCAFFOLD (multi-month seed — planted, not built)

- Gravity as substrate-volume coupling (F56). Since mass = substrate volume, gravity = the substrate curving in response to its own volume being occupied — the gravity-face of commitment-density ρ_{commit} . Assembles mass=volume + κ_{Bergman} + ρ_{commit} + the Friday G-chain + Einstein-from-commitment. Highest-value concrete target: derive G + the 8π from $\kappa_{\text{Bergman}} + \ell_B + \pi^{\{n_C\}}$ (would pin the G-chain’s open ℓ_B). Explicit falsifiers stated. Multi-month.

5. RETRACTED / PRUNED (recorded so we don't rebuild — the weekend's honesty)

- Substrate-Schur “P unifies muon + Λ ” — RETRACTED (Cal #259; F43/F44). The (1-P) two-meaning conflation. The unification was a conjecture resting on calling two things by one name.
- 81/40 vacuum match; $1432/g \approx 207$ — declined/retired (numerically close, no mechanism).
- Composite v0.5 additive ($m_\mu/m_e = 1575/8+81/8$) — RETIRED to the single-object T190 = $(24/\pi^2)^{C_2}$ (F50; double-counts under Reading (a); T190 fits 40× better and is Reading-(a)-safe).
- Lifetime $\propto 1/\text{Casimir}$ (“opposites explain half-lives”) — REFUTED (Elie 4016 + Grace, PDG data). Lifetime is mass/phase-space-dominated (Sargent m^5 , 0.3%); electron counterexample. Mass=volume untouched.
- ~200 cross-mass confirmations — pruned to 4 real (base-rate).
- Symmetric $\pi^2 = \pi^{\text{rank}}$ — refined to asymmetric (Grace); π^2 stays candidate, not a mirror of π^5 .

6. OPEN (the named next derivations — where work resumes)

- Spinor +1 rigor: the explicit winding integral yielding *exactly* $2\lambda_2$; Grace's base-sharing (baryon spinor shares the n_C base with the meson before the twist). [Keeper K249 + Lyra, near-term]
- The $n_C+2\lambda_2$ ladder test: do u/d hadrons sit at $7\pi^5$, $8\pi^5$? $a_1(1260)/f_2(1270)$ candidates. [Elie — falsifiable, the *real* Phase 3]
- Gravity step 3: $G + 8\pi$ from $\kappa_{\text{Bergman}} + \ell_B + \pi^{n_C}$, pinning ℓ_B . [Lyra, multi-month]
- “Why flat not invariant” (F54): the substrate-mechanism for mass = extensive flat volume. [Lyra]
- (1,1) bridge: $\rho_{\text{conf}} - \rho_{\text{SO}(5)}$ — derived-mechanism or convention-gap? Candidate-lead, weighs 0 (K231c). [Lyra/Keeper]
- Full R(k) theorem: explicit $a_k(n_C)$ closed form (a_k peaks then decays, no simple law — Elie 4014). [Elie's deferred session]
- CKM color²: Direction B (two-point, N_c^2 amplitude) confirmed at FRAME-WORK (F39/F50); the explicit FK color-tensored matrix element remains. [Lyra/Elie]
- Asymmetric dual- ρ structural pass + field-theory ρ -classification $\rightarrow$ Mass Gap grounding. [Grace]

7. The discipline record (why the survivors are load-bearing)

The weekend ran ~10 cross-CI catches, several against the team's own most-loved ideas, none defended: Cal #259 (unification), the lifetime refutation (both Elie + Grace), the base-rate prune, the symmetric- π refutation, the C_2 over-determination, the stale “muon-bulk” label, the 81/40 decline, my F36 A-lean, Keeper's K230 retraction, the +1-was-a-fit catch. The pattern (Cal #27): the

most elegant-feeling step is tested hardest, and the cheapest test runs first. What survived — mass=volume, $R(k)$ =curvature, the dual- ρ split, $\text{cells}=\text{n_C}+2\lambda_2$ — survived *because* it was tested. The π^5 and the spinor +1 both turned out to be “the plain question wearing a name tag”: $\pi^5 = \pi^{(\text{complex dimension})}$; +1 = the order of a Z_2 we’d already registered (T185).

8. Next-steps menu (for the team reconvene)

In rough priority, physics-lead read: 1. Cal cold-read the spine (F52–F57 + K238/K249) — check the foundation before building Tier 2 on it. *Recommended first.* 2. Consolidate + close Phase 3 (meson n_C + baryon $\text{n_C}+1$ into one note) + the spinor+1 rigor (Keeper K249) + the λ_2 -ladder test (Elie). Light, near-term. 3. Open gravity-as-volume forward work (F56 step 1: $\rho_{\text{commit}} \leftrightarrow \text{volume-occupation}$; toward step 3: $G + \ell_B$). The multi-month thread. 4. Tier 2 parallel: dual- ρ structural pass + field-theory ρ -classification (Grace).

9. Closure

We are at end-of-Phase-3-core. Phase 2 closed ($\pi^5 = \pi^{\{n_C\}}$); Phase 3’s two integers derived (meson n_C , baryon $\text{n_C}+1 = C_2$ via the Z_2 twist); the gravity thread seeded; the dual- ρ structure mapped. The load-bearing core is small and tested: mass is the substrate’s five-dimensionality made visible as weight, and the boson/fermion split is whether the winding band twists. What’s ahead is sharper than what was planned — a predictive mass-ladder ($\text{cells} = \text{n_C}+2\lambda_2$) to test, and gravity-as-volume to build — with the honest open derivations named in Sec. 6. Documented; standing for the team’s next-steps reconvene.

— Lyra, Sun 2026-06-07 12:10 EDT. Weekend state-of-knowledge consolidation. SPINE: mass = substrate volume ($\pi^{\{n_C\}}=\pi^5$, F52); how much you wind = mass, whether the band twists = boson/fermion. DERIVED: $\pi^5=\pi^{\{n_C\}}$; $m_p=6\pi^5=C_2\cdot\pi^{\{n_C\}}$; measure role-split (flat $\pi^{\{n_C\}}$ mass / FK $\pi^{\{9/2\}}$ spectral, gap 1/rank, F54); $R(k)=C(k,2)/\kappa_{\text{Bergman}}$ (F47/49); Phase 3 integers (meson n_C , baryon $\text{n_C}+1=C_2$ via Z_2 twist, derived 4 ways); 4 real cross-mass hadrons. CANDIDATE: mass=beyond-SO(4,2) conformal (F53, Casey #14); $\text{cells}=\text{n_C}+2\lambda_2$ spinor ladder (F57, predictive $\lambda_2=1\rightarrow 7\pi^5$); asymmetric dual- ρ (F49+Grace); Reading (a) (F44). SCAFFOLD: gravity-as-volume (F56, multi-month, target = $G+\ell_B$ from $\kappa_{\text{Bergman}}+\pi^{\{n_C\}}$). RETRACTED: substrate-Schur unification (Cal #259); 81/40; Composite v0.5 additive $\rightarrow$ T190 (F50); $\text{lifetime}\propto 1/\text{Casimir}$ (Elie+Grace); $\sim 200\rightarrow 4$ cross-mass; symmetric $\pi^2=\pi^{\text{rank}}\rightarrow$ asymmetric. OPEN: spinor +1 rigor + base-sharing (Keeper K249); λ_2 -ladder test (Elie); gravity $G+\ell_B$ (multi-month); why-flat (F54); (1,1) candidate-lead; full $R(k)$ a_k ; CKM Direction B explicit. DISCIPLINE: ~ 10 catches, none defended, survivors load-bearing because tested. Next: Cal cold-read spine first $\rightarrow$ Phase 3 consolidation + spinor rigor + λ_2 test $\rightarrow$ gravity forward + dual- ρ . End-of-Phase-3-core; ahead is sharper than planned.